2 Expected values of matrices

2.1 Expected value of variables

What we already know, for variables X, and Y, and constants a, and b:

$E(aX) = aE(X),\ E(X+Y) = E(X)+E(Y),\ E(aX+bY) = aE(X)+bE(Y),$

and for variables $X_1, \ldots, X_n$, and constants $a_1, \ldots, a_n$

$$E\left(\sum_{i=1}^{n} a_i X_i\right) = E(\underbrace{a_1 X_1 + \cdots + a_n X_n}_{\text{written out for clarity}}) = \sum_{i=1}^{n} a_i E(X_i)$$

2.2 Expected value of matrix

Let $\mathbf{X}$ be and $n \times m$ matrix of random variables. We define the **expected value of** the matrix $\mathbf{X}$ as the matrix

$$E(\mathbf{X}) = \begin{bmatrix} E(X_{11}) & E(X_{21}) & \cdots & E(X_{n1}) \\ E(X_{12}) & E(X_{22}) & \cdots & E(X_{n2}) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_{1m}) & E(X_{2m}) & \cdots & E(X_{nm}) \end{bmatrix}_{n\times m} \tag{2.1}$$

2.3 Expected value of scalar times matrix

For constant scalar a, by the definition in Eq. 2.1 and Section 1.2 Multiplication with scalar

$$E(a\mathbf{X}) = \begin{bmatrix} E(aX_{11}) & E(aX_{21}) & \cdots & E(aX_{n1}) \\ E(aX_{12}) & E(aX_{22}) & \cdots & E(aX_{n2}) \\ \vdots & \vdots & \ddots & \vdots \\ E(aX_{1m}) & E(aX_{2m}) & \cdots & E(aX_{nm}) \end{bmatrix}_{n\times m} \tag{2.2}$$

For each i and j, we have by properties of univariate expected values that

$$E(aX_{ij}) = aE(X_{ij}),$$

which mean that we can write Eq. 2.2 as

$$E(a\mathbf{X}) = aE(\mathbf{X}) \tag{2.3}$$

2.4 Expected value of sums of matrices

Let $\mathbf{X}$ and $\mathbf{Y}$ be two $n \times m$ matrices of random variables. From Section 1.1 Addition, we know that $\mathbf{X} + \mathbf{Y} = \mathbf{C}$. Applying the definition in Eq. 2.1 to $\mathbf{C}$ we have

$$E(\underbrace{\mathbf{X}+\mathbf{Y}}_{\mathbf{C}}) = \begin{bmatrix} E(X_{11}+Y_{11}) & E(X_{21}+Y_{21}) & \cdots & E(X_{n1}+Y_{n1}) \\ E(X_{12}+Y_{12}) & E(X_{22}+Y_{22}) & \cdots & E(X_{n2}+Y_{n2}) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_{1m}+Y_{1m}) & E(X_{2m}+Y_{2m}) & \cdots & E(X_{nm}+Y_{nm}) \end{bmatrix}_{n\times m}$$

For each i and j, we have by properties of univariate expected values that

$$E(X_{ij} + Y_{ij}) = E(X_{ij}) + E(Y_{ij}),$$

and we can write $E(\mathbf{X} + \mathbf{Y})$ as

$$\begin{aligned} E(\mathbf{X}+\mathbf{Y}) &= \begin{bmatrix} E(X_{11})+E(Y_{11}) & E(X_{21})+E(Y_{21}) & \cdots & E(X_{n1})+E(Y_{n1}) \\ E(X_{12})+E(Y_{12}) & E(X_{22})+E(Y_{22}) & \cdots & E(X_{n2})+E(Y_{n2}) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_{1m})+E(Y_{1m}) & E(X_{2m})+E(Y_{2m}) & \cdots & E(X_{nm})+E(Y_{nm}) \end{bmatrix}_{n\times m} \\ &= E(\mathbf{X}) + E(\mathbf{Y}) \end{aligned}$$

giving us the rule

$$E(\mathbf{X} + \mathbf{Y}) = E(\mathbf{X}) + E(\mathbf{Y}) \tag{2.4}$$

2.5 Expected value of linear (scalar) combinations of matrices

For constant scalars a and b, and $\mathbf{X}$ and $\mathbf{Y}$ that are $n \times m$ matrices of random variables

$$E(a\mathbf{X} + b\mathbf{Y}) = aE(\mathbf{X}) + bE(\mathbf{Y}) \tag{2.5}$$

This relies on the linearity of expectations (Section 2.1) and expectation of a variable times a scalar (Eq. 2.3):

$$E(a\mathbf{X} + b\mathbf{Y}) \underbrace{=}_{linear} E(a\mathbf{X}) + E(b\mathbf{Y}) \underbrace{=}_{scalar} aE(\mathbf{X}) + bE(\mathbf{Y})$$

2.6 Expected value of linear (vector) combinations of matrices

Let $\boldsymbol{x}$ be a $n \times 1$ vector of random variables, and $\boldsymbol{a}$ be a $n \times 1$ vector of constants.

$$
\begin{aligned}
E(\boldsymbol{x}\boldsymbol{a}^\top) \quad & \underbrace{=}_{\textit{outer prod}} \quad E\left(\begin{bmatrix} X_1 a_1 & X_1 a_2 & \cdots & X_1 a_n \\ X_2 a_1 & X_2 a_2 & \cdots & X_2 a_n \\ \vdots & \vdots & \ddots & \vdots \\ X_n a_1 & X_n a_2 & \cdots & X_n a_n \end{bmatrix}_{n\times n}\right) \\
& = \quad \begin{bmatrix} E(X_1)a_1 & E(X_1)a_2 & \cdots & E(X_1)a_n \\ E(X_2)a_1 & E(X_2)a_2 & \cdots & E(X_2)a_n \\ \vdots & \vdots & \ddots & \vdots \\ E(X_n)a_1 & E(X_n)a_2 & \cdots & E(X_n)a_n \end{bmatrix} \\
& = \quad \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{bmatrix} \begin{bmatrix} \text{---} & \boldsymbol{a} & \text{---} \end{bmatrix} \\
& = \quad E(\boldsymbol{x})\boldsymbol{a}^\top
\end{aligned}
$$

For the inner product from Section 1.5 and

$$
E(\boldsymbol{a}^\top \boldsymbol{x}) = \boldsymbol{a}^\top E(\boldsymbol{x}) \tag{2.6}
$$

Use the definition of inner product Eq (1.4)

$$
E(\boldsymbol{a}^\top \boldsymbol{x}) = \underbrace{E\left(\sum_{i=1}^{n} a_i X_i\right)}_{\text{def inner prod}} \overset{\text{Sec.2.1}}{=} \sum_{i=1}^{n} a_i E(X_i) = \boldsymbol{a}^\top E(\boldsymbol{x})
$$

By definition (Eq. 2.1) the expected value of the transpose

$$
E(\boldsymbol{x}^\top) = \begin{bmatrix} E(X_1) & \cdots & E(X_n) \end{bmatrix},
$$

so that

$$
E(\boldsymbol{x}^\top \boldsymbol{a}) = E(\boldsymbol{a}^\top \boldsymbol{x})
$$

For two random vectors $\boldsymbol{x}$ and $\boldsymbol{y}$

$$E(\boldsymbol{x}^\top \boldsymbol{y}) = E\left(\sum_{i=1}^{n} X_i Y_i\right) = \sum_{i=1}^{n} E(X_i Y_i)$$

and

$$E(\boldsymbol{x}\boldsymbol{y}^\top) = \begin{bmatrix} E(X_1Y_1) & E(X_1Y_2) & \cdots & E(X_1Y_n) \\ E(X_2Y_1) & E(X_2Y_2) & \cdots & E(X_2Y_n) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_nY_1) & E(X_nY_2) & \cdots & E(X_nY_n) \end{bmatrix}$$

If the matrix $\mathbf{A}$ has columns $\boldsymbol{a}_i$

$$E(\mathbf{A}^\top \boldsymbol{x}) = \begin{bmatrix} E(\boldsymbol{a}_1^\top \boldsymbol{x}) \\ \vdots \\ E(\boldsymbol{a}_p^\top \boldsymbol{x}) \end{bmatrix} = \mathbf{A}^\top E(\boldsymbol{x}) \tag{2.7}$$

consequently, for $n \times m$ matrix $\mathbf{X}$ of random variables and $n \times p$ matrix $\mathbf{A}$ of constants

$$E(\mathbf{A}^\top \mathbf{X}) = \mathbf{A}^\top E(\mathbf{X})$$

is a $p \times m$ matrix of expected values.